A carbon tax with planetary boundaries

Abstract

Human activities are threatening to push the Earth system beyond its "planetary boundaries", risking catastrophic and irreversible global environmental change [J. Rockström et al., Nature 461(7263), (2009)]. Action is urgently needed, yet well-intentioned policies designed to reduce pressure on a single boundary can lead, through global economic feedbacks, to aggravation of other pressures. In particular, the potential policy spillovers from a global carbon tax onto other critical Earth system processes has yet received little attention. To this end, we explore the global environmental effects of a carbon tax, beyond reducing carbon emissions. We find that the case for a global carbon tax is even stronger in a multi-boundary world, but that such a policy would create some important trade-offs among the nine boundary processes outlined by Rockström et al.. Further exploration of the policy space reveals that a carbon tax in combination with a biofuel policy could potentially alleviate all planetary pressures.

Main

The Earth has been in a remarkably steady state over the last 10,000 years, but human activities since the industrial revolution are now starting to threaten this balance. Rockström

and colleagues (*1, 2*) developed a list of nine Earth System Processes (ESPs) that are critical to maintaining a stable global environment, ranging from climate change to biodiversity loss to freshwater use. There are also nine corresponding "planetary boundaries" beyond which mankind cannot proceed without risking global environmental change with potentially catastrophic consequences for human civilization (*3*). Even as these planetary boundaries are gaining recognition among policymakers,[1] however, the complexity of the many interlocking processes can seem to present them with an unnavigable obstacle course.

The purpose of this paper is to assess the global environmental consequences of a global carbon tax in a multi-boundary world. Climate change mitigation is frequently viewed a matter of applying the breaks to greenhouse gas emissions, but a multi-boundary perspective alerts us to the possibility that it might inadvertently redirect economic activities in ways that exacerbate (or alleviate) other harmful environmental pressures. For instance, a carbon tax targeting fossil fuels would on its own most likely increase the demand for biofuels, stimulating production, and hence potentially becoming a driver of land-conversion and biodiversity loss (*4*). A specific policy that moves us away from one boundary could thus simultaneously push us towards another (*5*). A better analogy to the problem of policy-making with multiple planetary boundaries may therefore be that of "parallel parking", where the challenge is to simultaneously respect boundaries on all sides, ensuring a "safe operating space" for global societal development (*3*).

We present the result from a new global economic model that integrates multiple ESPs. Prior literature has focused on studying how policies affect a single ESP in isolation (*6, 7*), or developed large-scale computational models to provide counterfactual simulations for two closely related ESPs simultaneously, such as land-use/deforestation, or climate change/food

[1] They are cited repeatedly in the United Nations Environment Programme's world assessment reports GEO-5 and GEO-6, and the former UN secretary-general Ban Ki-moon endorsed the concept following a report from the High Level Panel on Global Sustainability. The draft document presented for world leaders at the Rio+20 Summit endorsed them. In 2015 the updated findings were presented in the World Economic Forum in Davos. It has also featured in several public media channels such as the New York times, Washington Post and The Economist to name a few.

security (*8–11*). Our model substantially expands the scope of analysis to include the whole of the planetary boundaries framework.[2] We provide a framework that is at once rich enough to let researchers investigate and discover the complex ways in which policies can interact with multiple ESPs, while also being simple enough to yield a qualitative understanding of the intended and unintended effects of global environmental policies. The model represents the quantitatively most important economic drivers of environmental change, is calibrated using data from the Global Trade Analysis Project (GTAP) (*12*), and links these economic sectors to the ESPs highlighted in the planetary boundaries framework. The model is designed to be transparent enough to enable independent assessment of the validity of the results (*13, 14*), while also being flexible enough to allow researchers to experiment with a wide range of policy options.

Our analysis reveals that a global carbon tax would, on its own, relieve pressure on most ESPs.[3] The case for a global carbon tax is therefore stronger in a multi-boundary world than when considering climate change in isolation. Such a tax would, however, increase the pressure on the freshwater and land-system processes. This is partly because higher fossil fuel prices make biofuel production more competitive. We therefore investigate the possibility of implementing a complementary biofuel policy as a counter measure and find that this combination of policies is able to ease all planetary pressures simultaneously.

[2]Importantly, the focus is not on whether the planetary boundaries will be transgressed or not but rather whether a policy targeted at a specific ESP has unintended outcomes, in terms of increasing or decreasing the pressure on other ESPs (i.e. the process moves towards higher risk of serious impacts or alternatively towards a safer operating space).

[3]Throughout the paper we will refer to a carbon tax as a fee imposed on the burning of carbon-based fossil fuel sources (e.g. coal, oil, gas). Although such a policy has not been implemented at a global level, several national examples exits (*15*). Recent studies also suggest that such a policy if combined with revenue recycling would likely receive public support (*16*). However, we note that this is a narrow definition of carbon taxes that could potentially be extended to cover emissions from all carbon based sources (e.g. biofuels, land-use change, etc.). Along these lines we also consider the removal of biofuel subsidies as a complementary policy.

Economic drivers of planetary pressures

The planetary boundaries can be viewed as a list of the greatest global environmental problems caused by and facing mankind. Since these environmental problems are driven by economic activities, the first step in understanding the nature of the problem is to identify the principal sources of anthropogenic pressure on each ESP, and the links between them. We thus research the literature detailing how specific activities in different economic sectors are affecting each ESP in the planetary boundary framework, and summarise our findings in table 1.[4]

As can be seen, agricultural activity is a substantial source of pressure on the ESPs in table 1.[5] Despite that it accounts for merely 4% of global economic output (*21*), it uses nearly 40% of the Earth's land surface and drives a vast majority of land conversion. It also contributes to roughly a quarter of global greenhouse gas emissions and accounts for over 90% of global freshwater, phosphorus and nitrogen use. Agriculture is thus a primary driver of freshwater over-consumption and biogeochemical loading. Apart from the agricultural sector, fossil fuel consumption is also a key source of pressure on several ESPs. This is due partly to its direct impact on climate change, aerosol loading and ocean acidification, but it also exerts pressure on other ESPs indirectly through economic channels. For example, biogeochemical flows are highly dependent on fossil fuels. From the foregoing discussion, it is evident that both agriculture and fossil fuel consumption, as well as their interaction, are vital components of any model that aims to capture how human activity exerts pressure on the ESPs.

[4]Our interpretations of the boundaries follow (*3*) with the exceptions of the biosphere integrity and novel entity boundaries. For these we use the definitions biodiversity loss and chemical pollution, from (*2*), since their interpretations are less ambiguous. Further details and references to these numbers can be found in SI (see section on "Drivers of planetary pressures").

[5]The centrality of agriculture has previously been pointed out in (*19, 20*).

Model description

The overall aim of the modeling exercise is thus to develop a framework for analyzing the effects of economic policies on all the planetary boundaries. The design is the conscious result of a trade-off between two overarching ambitions, i) include sufficient detail to capture the principal drivers of the pressures on the planetary boundaries; and ii) keep it simple enough to allow for an intuitive understanding of the key economic drivers. We follow standard practice in economic modeling by characterizing a decentralized market structure, with a multitude of economic actors that simultaneously maximize their individual objectives, resulting in competition for the limited set of resources, and thus often referred to as a competitive equilibrium. We thus want to expand the scope of models aimed at studying single boundaries, such as the DICE model for climate change, while keeping the model significantly more transparent than models incorporating multiple planetary boundaries such as the IMAGE model (*14*).

Figure 1 provides a schematic outline of the model. Guided by the findings summarized in table 1, the model characterizes production choices in the key economic sectors: agriculture, biofuels, timber production, fertilizer production, phosphate extraction, water extraction, manufacturing, energy services, fossil-fuel production, renewable energy and fisheries. In total, these sectors account for more than 90% of the drivers for the majority of the planetary pressures. Many of the sectors are economically linked, both through the inputs they require and outputs they produce. These linkages allow us to study many critical trade-offs, for instance the allocation of land across agriculture and timber production (inline with e.g. (*22*)) or agricultural output, which can be used as either food or biofuel (similar to (*23*)).

The model details are outlined in the methods section and supplementary information (SI). Two broad features of our modelling approach are however worth pointing out. First, compared to standard solution methods used in most IAMs, which involves completely solving for the equilibrium, we instead look at the solution using comparative static methods, an approach that

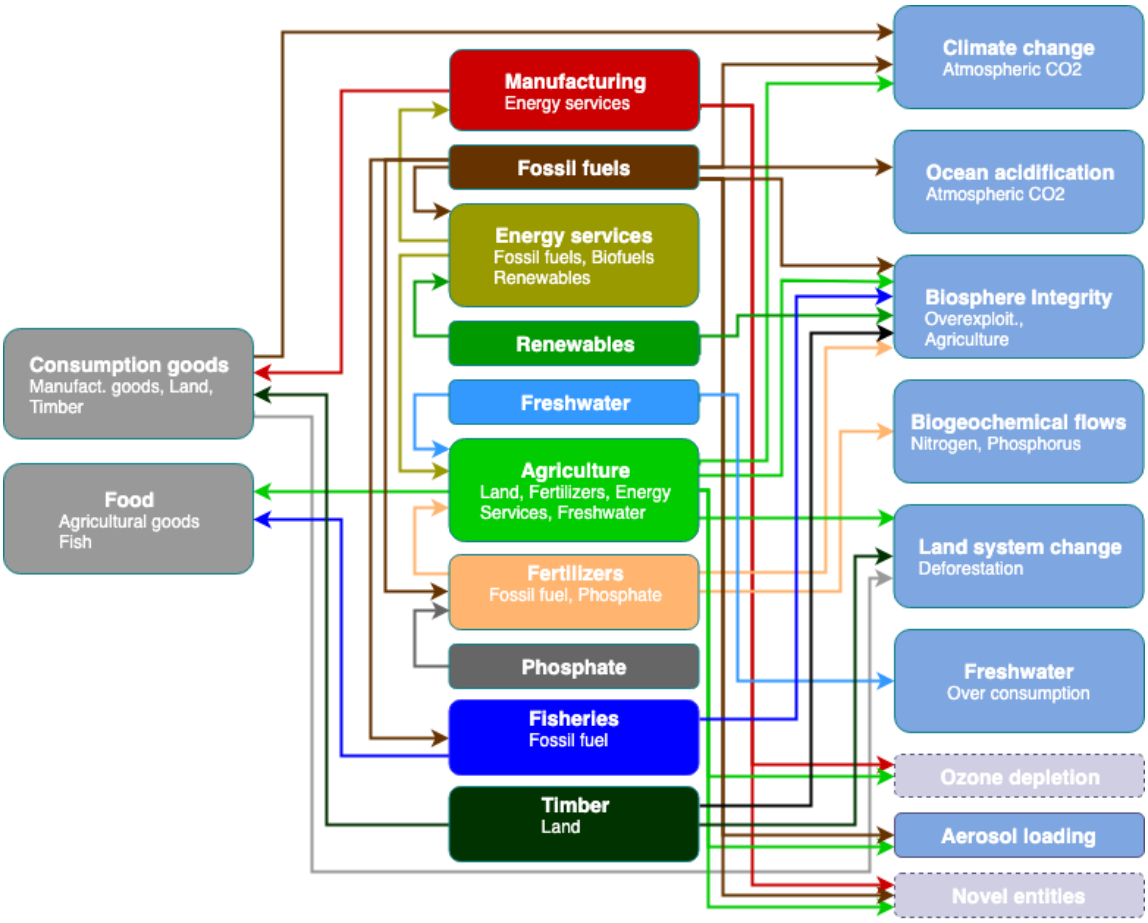

Figure 1: **Schematic of the integrated economic-planetary boundaries model.** The above schematic gives an overview of the direct links existing in the model. The model is built in three layers (columns): consumption, production, and earth system processes. The arrows indicate the direction of the economic inputs and outputs and which planetary processes they have an impact on. With exception of ozone depletion and chemical pollution the impact of the two policy scenarios we consider are quantitatively assessed.

can be characterized as a linear approximation around the equilibrium outcome. Interventions like taxes and quotas can thus be represented as perturbations to the equilibrium, making it easy to trace out how their effects propagate through the entire economy. Apart from increased transparency, this approach also has the advantage that the model parameters are easier to interpret. In total, three broad sets of parameter estimates need to be assessed, i) elasticities (demand, supply and substitution), ii) expenditure shares of the economic sectors and iii) quantity shares (see

Methods). Together, these features make the model and results derived easier to understand and critically assess in comparison to the large-scale IAMs using standard solution methods (*13*).

In addition, our focus is on providing a qualitative understanding of the central processes and interactions involved. Hence, our modelling framework does not at present incorporate dynamic aspects of the problem. Also, the model does not include feedbacks from the planetary pressures back onto the economy and human welfare. These feedbacks, although important, are unfortunately not understood nearly well enough to inform this kind of modelling exercise, and are thus left for future research. The effects on the planetary pressures are however computed as the net effects of the policy induced changes in the various economic activities. In summary, our analysis is intended to capture the effects of changes in the economic environment in the short- to medium-term (perhaps 5-10 years). A longer-term analysis would need to take additional aspects, such as technological change and dynamic feedback already referred to, into account. One should be careful, therefore, in applying the framework developed here to long-run decision problems where dynamic feedbacks are likely to be important.

Model parameterization

Among the three broad set of parameter estimates described above, the majority of the estimates needed can be expressed in terms of expenditure and quantity shares of goods going to different uses (see methods). Detailed sectoral databases, such as the one underlying the GTAP model (*12*) – widely used in economic research – allow us to derive a coherent set of parameter values. Most of the model's 46 parameters are consequently derived from the GTAP data, while a few parameters (e.g. supply and substitution elasticities) are chosen to match earlier empirical studies. A complete list of parameters and their sources can be found in the methods section.

A global carbon tax

We now consider the policy experiment involving a marginal increase in the tax rate on fossil fuels in order to study its effect on the equilibrium outcome. As described above, this is implemented as a linear approximation around the equilibrium which is scaled to a one percentage point increase in the tax rate. The scaling up of this marginal effect is an approximation which holds to the extent that the linear approximation remains valid (which presumably holds for small changes in the tax rate).

The direct effect of imposing this increase in the tax on fossil fuel use is clearly to increase the (after tax) price and hence reduce demand for it. Our analysis shows that a one percentage point increase in the global carbon tax rate would reduce annual global CO_2 emissions by 0.25% (equivalent to 0.11 $GtCO_2$ yr^{-1}). This would ameliorate the pressure on both climate change and ocean acidification (depicted as arrows at the top and bottom of figure 2, pointing inward towards the safe operating space). This change does not come from a single sector, but is rather the net effect of changes in fossil fuel use in the production of energy services, fertilizer, land use change, etc. CO_2 emissions from fossil fuel use decrease in most sectors as a result of the tax, while emissions from land use change increase due to substitution towards land in agriculture and increased biofuel production. Similar direct impacts occur with aerosol loading. Here the net effect (measured in terms of aerosol optical depth) is reduced following the reduction in the release atmospheric particles from fossil fuel burning.

As producers and consumers of fossil fuel react to changing prices, this leads to several second-order effects resulting from the carbon tax. To start with, an increase in the carbon tax increases the cost of the nitrogen component of fertilizers, since nitrogen fixing uses a very fossil-fuel intensive industrial process. Since nitrogen has a high degree of complementarity with phosphate in the production of fertilizers (most fertilizers are sold as multi-nutrient mixtures), demand for phosphate also decreases, reducing the overall pressure on the biogeochem-

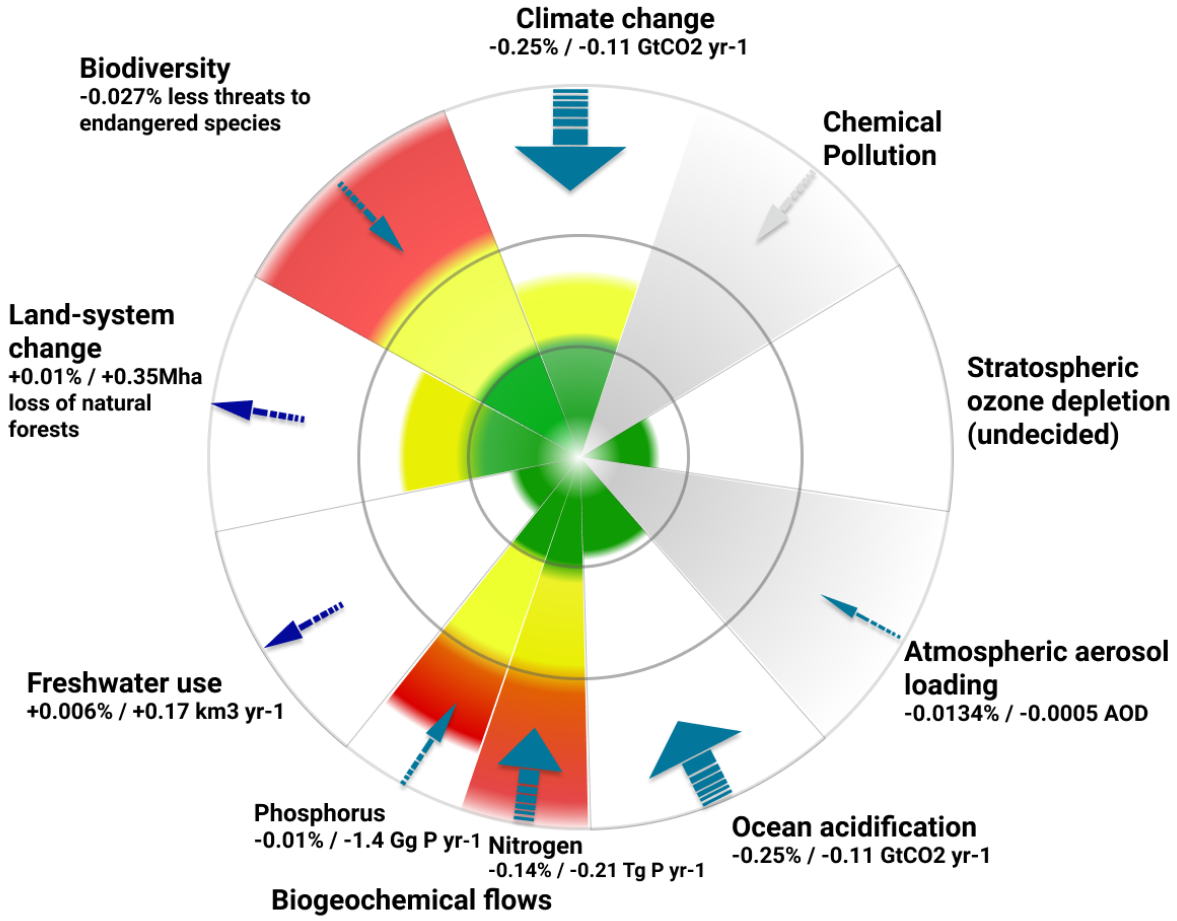

Figure 2: **Changes in planetary pressures as a result of a one percentage point increase in the carbon tax.** The figure shows a modification of the original planetary boundary figure from (*1, 3*), where the effects on each individual ESP, from increasing the carbon tax rate by one percentage point in our integrated model of the global economy and the ESPs, have been added. The direction of the arrows indicate increasing or decreasing pressure, while the width of the arrows are indicative to the magnitude of change. For chemical pollution and stratospheric ozone depletion we only derived the qualitative direction of change.

ical flows. Furthermore, a carbon tax turns out to reduce pressure on our adopted measure of biodiversity. This is partially due to reduced activity in key sectors such as agriculture and fisheries, in which fossil fuel is an important input. The total effect on biodiversity thus comes from

a combination of a number of different effects, some of them negative others positive (see SI, section “Determining how policy-induced changes impact the planetary pressures”, for details on how the net effect is derived). Chemical pollution and stratospheric ozone depletion are only qualitatively assessed based on their relationship to the model variables. If all related model variables move in a direction that decreases the pressure, we draw the conclusion that the net effect is reduced pressure. This turns out to be the case for chemical pollution. For stratospheric ozone depletion, however, the effects go in different directions and we cannot determine the net direction (see methods/SI for details).

A carbon tax will also increase the pressure on land-system change, a net result of several opposing effects. First, a higher relative price of non-land inputs (energy and fertilizer) will encourage substitution towards greater land use in the agricultural sector. At the same time, the higher price of fossil fuel raises the relative price of the manufactured good compared to e.g. timber and recreation. The demand for non-agricultural uses of land thus counteracts the increase in demand from agriculture. Together with land conversion costs, this results in a small overall increase in land use in the agricultural and timber sectors, at the expense of natural land. Finally, a carbon tax leads to a small increase in freshwater use. In this case, the increased freshwater use is due primarily to substitution away from more expensive agricultural inputs.

An increase in the global carbon tax thus reduces pressure on many ESPs besides climate change. The argument for a carbon tax is typically made considering only climate change. Our analysis suggests that in a richer framework, that considers multiple planetary boundaries, the case for a carbon tax is in some ways even stronger. This richer model does alert us to certain risks, too, and next we investigate whether a complementary biofuel policy could avert these dangers and move us more firmly towards the safe operating space.

Reduction of biofuel subsidies

A global carbon tax tends to put pressure on the land-system and freshwater ESPs mainly because it increases demand for agricultural output. Delving deeper into the sectoral linkages, one can see that this is not chiefly demand for food, but rather demand for biofuels (which is a substitute for fossil fuel). In order to ease the remaining planetary pressures, it is thus likely that we need to complement the carbon tax with some additional policy limiting the demand for biofuels.

We consider the effect of a complementary reduction of biofuel subsidies (implemented as an increase in a biofuel tax). Biofuel production is currently heavily subsidised in large parts of the world (*24*). The question is whether it would be prudent to scale back these subsidies in a world with a higher carbon tax. Figure 3 summarizes the net changes in our model resulting from this two-pronged tax policy: a one percentage point increase in the carbon tax rate and a one percentage point reduction of the subsidy rate for biofuels. As can be seen this combination of policies can ameliorate the pressures on all ESPs.

Our results thus highlight that biofuel production not only has a negative climate effect from increased land use, but also a negative impact in the form of increased demand for fertilizer and freshwater, thereby increasing the pressure on both the biogeochemical flows and freshwater systems.[6] Importantly, this result does not necessarily imply that a biofuels subsidy is a bad idea in the absence of a carbon tax. Such a policy would reduce some pressures while increasing others. Furthermore, it should be noted that we have biofuel production modeled as using the same inputs as food agriculture. Other types of biofuels, that are produced differently, would perhaps fit better into our renewables-category.

[6]Similar result have been found by (*25*) in an assessment of the consequences of a large-scale deployment of bioenergy with carbon capture and storage as a measure for mitigating climate change.

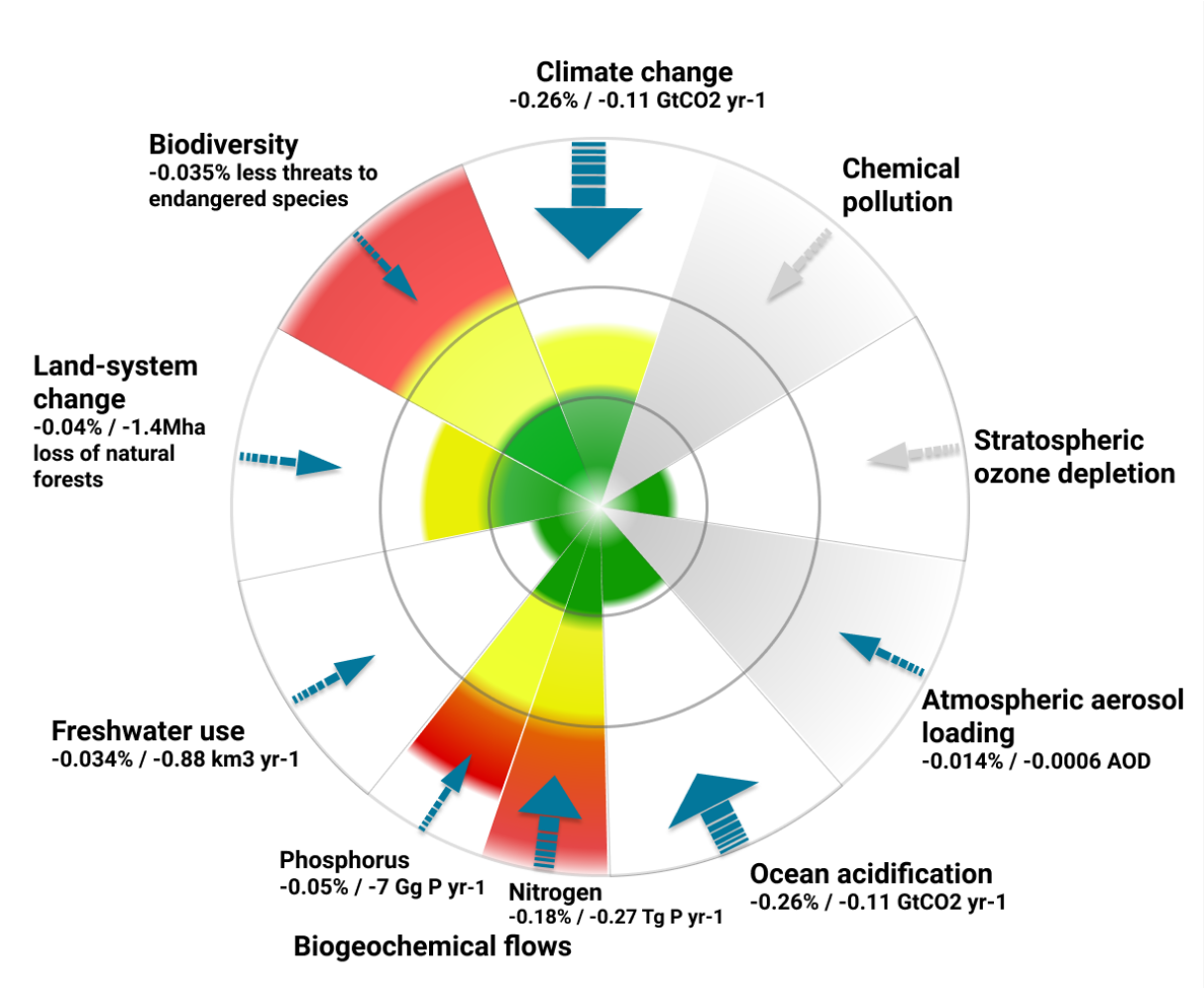

Figure 3: **Change in planetary pressures as a result of a one percentage point increase in the tax on carbon and a one percentage point reduction of biofuel subsidies.** The figure shows a modification of the original planetary boundary figure from (*1, 3*), where the effects on each individual ESP, from increasing the tax rate on carbon by one percentage point and reducing biofuel subsidies by one percentage point in our integrated model of the global economy and the ESPs, have been added. Interpretation is otherwise equivalent to figure 2.

Sensitivity analysis

To assess the robustness of the results produced in our model, we identify key parameters for which there is more uncertainty and choose lower and upper bound values for them. We then

solve the model for all possible combinations of these lower and upper bounds, recording the maximum and minimum predicted changes for all model variables.

In the case with a carbon tax as the only policy, the signs of the changes in planetary pressures are mostly unaffected, though there are a few notable sign changes. In some extreme scenarios, land use in agriculture and freshwater use may both decrease, and in rare cases we also observe an increase in phosphate use. Overall, our sensitivity analysis suggests that, if anything, the outcome distribution tends to be skewed toward positive rather than negative outcomes for planetary pressures.

In the case with both a carbon tax and a complementary biofuel subsidy reduction, the results are more robust. There is no reversal in the sign of the net effects. The only change of any relevance for our analysis is that we can get an increase in food production from agriculture. The details of the sensitivity analysis are reported in table **??** (supplementary information) for the carbon tax with and without the accompanying biofuel policy.

Conclusions

Carbon taxes are typically justified purely on the basis that they will help mitigate climate change. The planetary boundaries framework developed by Rockström and colleagues (*1*) begs the question as to whether carbon taxes could have unintended destabilizing effects. Here we present an integrated analysis showing that a higher global carbon tax would be expected to single-handedly reduce almost all of the planetary pressures. Some of the effects are direct and easy to anticipate, since fossil fuel consumption exerts an important pressure on several ESPs, including climate change and ocean acidification. Other consequences are more indirect, such as a reduction in nutrient-loading, which derives from the importance of fossil fuels in nitrogen production and complementarity between phosphate and nitrogen in fertilizer production.

A carbon tax can also indirectly increase other planetary pressures, though, especially

through increasing demand for agricultural land when producing more biofuel as a reaction to higher fossil fuel prices. We show that a complementary policy of scaling back biofuel subsidies as the carbon tax is increased would help avoid these negative effects on land and freshwater use. This combination of policies provide a way to reduce pressure on all of the earth system processes we have considered, and help navigate us towards a safe operating space for humanity.

The overarching goal of this paper has not been to calculate optimal policy paths for specific human activities such as carbon emissions or biofuel production. Instead it has been to present a transparent framework that allows for a policy analysis where direct and indirect effects on multiple earth system processes can be considered simultaneously. To this end we opted for a tractable static model. Our analysis clearly shows that indeed these interacting effects can be quite substantial and should not be ignored. While we considered two specific policies in this paper, the model can very easily be adapted to study the effects of a wide variety of policies. For future research we welcome extensions of our framework, including for example time dynamics, uncertainties, and welfare analyses.

Methods

The results of this paper are derived from a model which is built around the economic sectors outlined as the most important drivers of planetary pressures in table 1. This includes production sectors that have an important direct effect on the ESPs or that have important links to such sectors. They may be linked by using output from such sectors as inputs, providing inputs to such sectors, competing for inputs with such sectors or providing outputs that serve as substitutes for the output from those sectors. The resulting set of included production sectors are: agriculture (producing food and biofuel), energy services, fossil-fuel extraction, renewable energy (other than biofuel), fertilizer production, phosphate extraction, water supply, fisheries and

industrial manufacturing. The demand for final consumption goods is derived from maximization of households' utility. Since we have economic policies in the model, we are implicitly assuming some government entity that imposes these policies, but since we consider the policies exogenous (not, e.g., determined to optimize some objective) we do not explicitly model the government.

We solve the model as a competitive equilibrium where we assume that all agents maximize their respective objectives while taking prices as given (prices are given from the perspective of the individual agent, but are endogenously determined by aggregate supply and demand). We then analyze changes in the endogenously determined model variables in response to an assumed exogenous change in economic policy.

In the model, competition for resources thus leads to a number of important trade-offs. These arise from three main sources, i) alternative uses of the output of a sector (e.g. output from the agricultural sector can be used as food or biofuels), ii) sectors competing for the use of inputs (e.g. land can be used for agriculture, forestry or maintained as undisturbed natural land), or iii) from inputs being substitutable or complementary in production or consumption (e.g. nitrogen and phosphorus preferably being used in fixed proportions).

Model components

The production sectors are modeled either by an explicit production function or by a production cost function. A production function is specified for agriculture, energy services, fertilizer production, fisheries, timber production and industrial manufacturing sectors since their factor inputs are directly connected to one or more ESPs (see section on "Economic drivers of planetary pressures"), thus making their input substitutability important. For all sectors except agriculture, we use one level Constant Elasticity of Substitution (CES) functions. For agriculture we use a nested CES function (see below). Sectors whose production processes are of less

importance, are represented by an exponential production cost function. These sectors include phosphate, water, fossil fuel and renewable energy. Also, in many sectors, certain inputs e.g. labour and capital, are economically important but their explicit modeling is not directly relevant for our analysis (i.e. of negligible importance to the ESPs). To account for these inputs, we include an aggregate input, "other inputs", in all production sectors except energy services and assume that these are supplied with a given sector-specific price elasticity of supply.

We will now present the model sectors in more detail. A list of model quantities, their prices and uses can be found in Table 2 (different uses of a quantity are denoted by subscripts).

The agricultural sector uses inputs land (L_A), fertilizers (P), water (W), energy services ($\mathcal{E}_A$) and other inputs (M_A) as inputs to produce output that can be used for food or biofuels. Producers maximize their profit, taking prices as given. Their profit maximization problem is

$$\max_{L_A,P,W,\mathcal{E}_A,M_A} p_A A\left(L_A, P, W, \mathcal{E}_A, M_A\right) - p_L c_A(L_A) L_A - p_P P - p_W W - p_{\mathcal{E}} \mathcal{E}_A - p_{M_A} M_A, \quad (1)$$

where $c_A(L_A)$ captures the cost of converting land to agricultural land. The agricultural production function is a CES function between land and non-land inputs, where non-land inputs are aggregated using a CES function.

The energy-services sector combines energy from different sources into a bundle of energy services ($\mathcal{E}$). The different sources are biofuels (A_B), fossil fuels ($E_{\mathcal{E}}$) and renewables (R). The producers in this sector solve the profit maximization problem

$$\max_{A_B,E_{\mathcal{E}},R} p_{\mathcal{E}} \mathcal{E}(A_B, E_{\mathcal{E}}, R) - p_A A_B - p_E E_{\mathcal{E}} - p_R R. \quad (2)$$

We model production of fertilizers (P) as using fossil fuel (E_P), phosphate ($\mathcal{P}$) and other inputs (M_P). The use of fossil fuel is intended to capture the fossil-fuel (more specifically natural-gas) intensive production of the nitrogen component of fertilizers. We thus treat fossil fuel use in fertilizer production as a proxy for nitrogen. The profit maximization problem of

fertilizer producers is

$$\max_{E_P,\mathcal{P},M_P} p_P P\left(E_P,\mathcal{P},M_P\right) - p_E E_P - p_{\mathcal{P}}\mathcal{P} - p_{M_P} M_P. \tag{3}$$

For timber production (T) we only consider the input land (L_T) and other inputs (M_T). The producers then solve the maximization problem

$$\max_{L_T,M_T} p_T T(L_T,M_T) - p_L c_T(L_T) L_T - p_{M_T} M_T, \tag{4}$$

where c_T is a cost of converting (e.g., clearing) land for forestry.

Industrial manufacturing (Y) requires energy ($\mathcal{E}_Y$) and other inputs (M_Y). The maximization problem of the representative producer is

$$\max p_Y Y\left(\mathcal{E}_Y, M_Y\right) - p_{\mathcal{E}}\mathcal{E}_Y - p_{M_Y} M_Y. \tag{5}$$

The fisheries sector uses inputs fossil fuel E_F and other inputs M_F. The producers solve maximization problem

$$\max_{E_F,M_F} p_F F(E_F,M_F) - p_E E_F - p_{M_F} M_F. \tag{6}$$

Extraction of fossil fuel (E) is modelled by assuming a gross extraction cost (g_E) that increases with increased extraction ($g_E(E)$ thus gives the total cost of extracting quantity E). We assume that the tax on fossil fuels (a percentage tax τ_E) is paid by the firms that extract and sell it. The profit maximization problems of the extraction firms is then

$$\max_{E} \frac{p_E}{1+\tau_E} E - g_E(E). \tag{7}$$

The sectors phosphate ($\mathcal{P}$), water (W), renewable energy (R) and the other inputs (M_A, M_F, M_P, M_T and M_Y) are similarly represented by a production or extraction cost and the profit-maximization problem of the producers are given by

$$\max_{X} p_X X - g_X(X) \text{ for } X \in \{\mathcal{P}, W, R, M_A, M_F, M_P, M_T, M_Y\}. \tag{8}$$

We have now described the maximization problems underlying decisions made by all producers. The representative household also solves a maximization problem, maximizing the utility derived from consumption. The households' preferences are represented by utility function U and the utility-maximization problem, subject to the income being I, is given by

$$\max_{A_{\mathcal{F}},F,Y,L_U,T} U\left(\mathcal{F}\left(A_{\mathcal{F}},F\right),\tilde{\mathcal{F}}\left(Y,L_U,T\right)\right) \tag{9}$$
$$\text{s.t. } p_A A_{\mathcal{F}} + p_Y Y + p_F F + p_L L_U + p_T T \leq I.$$

This specification has divided consumption into two levels. While this division is not necessary at this level of generality, it clarifies the assumed substitutabilities between goods. We assume greater substitutability within than between categories. The upper level consists of food ($\mathcal{F}$) and non-food ($\tilde{\mathcal{F}}$) goods, with the former category consisting of food from agriculture and from fisheries, and the latter of manufactured goods, natural land and timber. The inclusion of natural land is intended to capture various ways in which households' demand for natural lands lead to land being kept from other uses, e.g., preservation of land as national parks. We assume that timber is consumed directly by the households.

This completes the description of the modelling of all decision-making agents in the model. In addition to conditions derived from these maximization problems, we must also specify market-clearing conditions that make sure that supplied and demanded quantities add up.

For land (L), the total supply is assumed to be fixed:

$$L = L_A + L_T + L_U. \tag{10}$$

The remaining market-clearing conditions are for agricultural production

$$A = A_{\mathcal{F}} + A_B, \tag{11}$$

fossil fuel

$$E = E_{\mathcal{E}} + E_F + E_P \tag{12}$$

and energy services

$$\mathcal{E} = \mathcal{E}_A + \mathcal{E}_Y. \tag{13}$$

In summary, production functions, market-clearing conditions, budget constraints and first-order conditions from the maximization problems of representative agents provide us with 41 equilibrium conditions pinning down the 41 endogenous prices and quantities. The full set of equilibrium conditions are available in the SI.

Solution Approach

We note a few features of our model, some of which have already been mentioned: there are no explicit externalities; policies are applied exogenously; all sectors are assumed to be competitive; market clearing determines the equilibrium. In this context, we can work with the decentralized equilibrium, which may be analyzed by considering the first order conditions. In our model, there are 41 unknown prices and quantities in the model, determined by 41 equilibrium conditions. Being exogenous, policies represent parameters that are known in advance; denote a generic "policy" pertaining to any one ESP by τ. Let X_i denote the generic i^{th} variable, an endogenous price or quantity. The jth equilibrium condition can then generally be written as:

$$G_j\left(X_1, \ldots, X_{41}; \tau\right) = 0. \tag{14}$$

This system of equations implicitly define all resulting equilibrium quantities and prices as functions of the policy i.e. $X_i = X_i\left(\tau\right)$.

There are now two solution approaches: the first is to solve the set of resulting non-linear equations (and thereby obtain all the equilibrium values); the second is to trace out marginal changes in the equilibrium values in response to a change in the policy, τ. The latter approach can be illustrated by considering the total derivative of the equilibrium conditions with respect

to the policy. This leads to a system of equations, with the jth equation being

$$\sum_{i}^{41} \left[\frac{X_i}{G_j} \frac{\partial G_j}{\partial X_i} \hat{X}_i \right] = -\frac{1}{G_j} \frac{\partial G_j}{\partial \tau}, \tag{15}$$

where

$$\hat{X}_i \equiv \frac{1}{X_i} \frac{dX_i}{d\tau} \tag{16}$$

is the relative change in variable X_i.[7] This is a system of 41 equations in 41 unknowns, the $\hat{X}_i$, and is most useful because of linearity in the unknowns. Indeed this approach can be viewed as linear approximation of the equilibrium response to a change in the policy parameter. The required empirical parameter values needed for numerical computations are fewer, easier to find , and easier to interpret. Furthermore, if considering changes in other parameters of the model (e.g., changes in other policies) only the right-hand side of (15) needs to be changed.

Data and parametrization

We parameterize the model based partly on data extracted directly from the widely-used GTAP database, described below, and partly on empirical estimates from various sources in the literature. As described above, we mainly need three types of values: quantity shares, expenditure shares and elasticities of various kinds. In total we need about 50 empirical values. In the following section we provide tables with parameter values and their sources.

Quantity shares

By quantity share Q_{X,X_Z} we mean the share of total quantity X used in a specific sector Z. The full set of values including their sources are given in table 3. The exceptions are the quantity

[7] These can be interpreted as a linear approximation of the percentage change in the variable induced by a one percentage point increase in the fossil fuel tax. Assume, for instance, that we get $\hat{X}_i = 2$ and consider a one percentage point increase in the tax rate, $\Delta\tau_E = 0.01$. We would then get $\frac{1}{Q_i}\Delta Q_i \approx \hat{Q}_i \Delta\tau_E = 0.02$. Hence, a one percentage point increase in the tax induces a two percent increase in the quantity.

shares of fossil fuel going to different sectors and the share of agricultural production going to food or biofuel. These were derived as follows.

Total energy consumption in 2011 was 12225 Mtoe (*26*). Out of this, 10624 Mtoe came from fossil fuel related sources. Fertilizer production uses about 1.2% of total energy supply and almost all of this comes from fossil fuels (*27*). Hence we assume that the share of fossil fuels going to fertilizer production is $\frac{12225}{10624} \times 1.2\% \approx 1.4\%$. For fisheries production we assume a global fuel consumption of 40 billion litre's of fuel (*28*). Assuming that this is mostly diesel, this corresponds to 40 Mtoe of fossil fuel or $\frac{40}{10624} \approx 0.4\%$ of total fossil fuel use. Finally we assume the remaining fossil fuels are used in energy production i.e. 98.2%.

Total biofuel production in 2007 was 36.1 Mtoe (*29*). Food consumption in 2005/2007 was 2772 kcal per person and day while the world population was 6569 million people (*30*). This adds up to a total food consumption of $2772 \times 365 \times 6569 \times 10^6 \approx 6.65 \times 10^{15}$ kcal. Using a conversion factor of 1.001×10^7, this corresponds to 665 Mtoe. The share going to biofuels is thus $\frac{36.1}{36.1+665} \approx 0.0515$

Expenditure shares

Another type of object that occurs in our equilibrium conditions are the expenditure shares. The expenditure share Γ_X^Z of input X in sector Z is the share of total spending on inputs in sector Z that goes to X. To pin down these at the global level, we employed the GTAP database (*12*). More specifically, we used the GTAP data set corresponding to the year 2014, for 141 countries and 57 sectors. The GTAP database is a unique global economic data set constructed by collating and reconciling data on national input output tables, international trade, production, consumption and macro-economic data sets from various international data sources. This has further been extended by (*33*) to include renewable energy commodities, based on several energy data sources including the International Energy Agency (IEA) data set and the World Bank

data set. Furthermore, (*34*) extended this even further to include water as an endowment, used in both agricultural and other sectors. Finally, we have a data set in which we can derive the shares of labor, capital, land, water, and several other inputs in producing all commodities. Some inputs such as fertilizers are not separately identified in this data set, but they are subsumed in broader GTAP sectors such as chemicals, rubber and plastics. Therefore, we make broad reasonable assumptions to derive the shares of such granular-level inputs; for example, we assume that most of agricultural consumption of output from the GTAP sectors chemicals, rubber and plastics are fertilizers and pesticides. For all production sectors except energy services, we assign the residual expenditure share, remaining when all inputs of direct interest have been accounted for, to "other inputs" M. The results are summarized in Table 4.

Agriculture Our agricultural production function distinguishes between land and non-land inputs (with "other inputs" in the non-land category). The expenditure share of land is 19.2%. The expenditure shares of fertilizers, water, energy and other inputs are 6.43%, 1.93%, 3.33% and 71.1% respectively. Their respective shares out of non-land inputs are their total shares divided by the total non-land share. This means that $\Gamma^{A}_{L_A} = 0.192$, $\Gamma^{A}_{\tilde{L}_A} = 0.808$, $\Gamma^{\tilde{L}_A}_{P} = \frac{0.0643}{0.808} = 0.0796$, $\Gamma^{\tilde{L}_A}_{W} = \frac{0.0193}{0.808} = 0.0239$, $\Gamma^{\tilde{L}_A}_{\mathcal{E}_A} = \frac{0.0643}{0.808} = 0.0412$ and $\Gamma^{\tilde{L}_A}_{M_A} = \frac{0.711}{0.808} = 0.880$.

Energy services The expenditure shares of biofuels, fossil fuels and renewables are 0.37%, 94.33% and 5.30% respectively. That is $\Gamma^{\mathcal{E}}_{A_B} = 0.0037$, $\Gamma^{\mathcal{E}}_{E_{\mathcal{E}}} = 0.9433$ and $\Gamma^{\mathcal{E}}_{R} = 0.0530$.

Utility The expenditure shares of food from agriculture, fish, manufactured goods, recreational land use and timber are 11.93%, 0.42%, 86.86%, 0.15% and 0.65%. This gives expenditure share of food $\Gamma^{U}_{\mathcal{F}} = 0.1235$ and expenditure share of non-food goods $\Gamma^{U}_{\tilde{\mathcal{F}}} = 0.8765$. The within-category expenditure shares are $\Gamma^{\mathcal{F}}_{A_{\mathcal{F}}} = \frac{11.93}{12.35} = 0.9660$, $\Gamma^{\mathcal{F}}_{F} = \frac{0.42}{12.35} = 0.0340$, $\Gamma^{\tilde{\mathcal{F}}}_{Y} = \frac{86.86}{87.65} = 0.9910$, $\Gamma^{\tilde{\mathcal{F}}}_{L_U} = \frac{0.15}{87.65} = 0.0.001711$ and $\Gamma^{\tilde{\mathcal{F}}}_{T} = \frac{0.65}{87.65} = 0.007416$.

Timber The expenditure shares of land and other inputs are 37.48% and 62.52% respectively. That is $\Gamma^T_{L_T} = 0.3748$ and $\Gamma^T_{M_T} = 0.6252$.

Composite good The expenditure shares of energy services and other inputs are 6.38% and 93.62% respectively. That is, $\Gamma^Y_{\mathcal{E}_Y} = 0.0638$ and $\Gamma^Y_{M_Y} = 0.9362$.

Fertilizers The expenditure share of energy is 10.95%. The factor share of phosphate is assumed to be a share $\xi_{\mathcal{P}} = 0.5$ out of the factor share of non energy intermediates 62.53%. That is $\Gamma^P_{E_P} = 0.1095$ and $\Gamma^P_{\mathcal{P}} = 0.5 * 0.6253 = 0.3127$. this leaves the expenditure share or other inputs as $\Gamma^P_{M_P} = 0.5778$

Elasticities

Finally, we need estimates of the elasticity of substitution, price elasticity of supply and elasticities of conversion cost estimates for several model quantities. For the majority of parameters we were able to track down estimates from the literature which are presented together with their corresponding reference in Table 5. Where the uncertainty in the estimates where high we employed a wide band for the sensitivity analysis. The parameters that are varied in the sensitivity analysis are indicated as [min, max, mean] with mean being the baseline values.

Numeric results

The full sets of changes in our model quantities and prices resulting from the two policies are presented in Table 6.

Determining how policy-induced changes impact planetary pressures

In this section we describe the mapping from changes in model variables to effects on ESPs. For the model variables freshwater (W), natural land-use (L_U), phosphate ($\mathcal{P}$) and nitrogen

(assumed to be proportional to fossil fuel use in fertilizer production E_P), there is a simple one-to-one mapping with model variables. For climate change, ocean acidification, biodiversity loss and aerosol loading, however, the mapping is more complicated. For climate change and ocean acidification we measure the change in pressure on both ESPs as the net change in CO_2 emissions. For biosphere integrity we measure changes in pressure as a change in threats to endangered species (more details on this below). We measure aerosol loading as changes in aerosol optical depth. For chemical pollution and ozone depletion we map pressures to contributing sectors, but do not make any quantitative analysis of the net effects.

Climate change and ocean acidification

Climate change and ocean acidification are both driven by carbon emissions and we use these emissions as the measure of the pressures associated with these boundaries. To translate changes in model variables into changes in emissions we use data from (*45*) and (*46*). From the figure on page 2 of (*45*) we get the percentage contribution of carbon dioxide emissions per sector outlined in the report. Using these percentages we can thus recover the amount of actual carbon emissions in gigaton carbondioxide ($GtCO_2$) per year connected to a specific variable in our model.

Using this approach, we start by looking at the energy related emissions that, according to (*45*), account for a total of 66.5%. Multiplying by the aggregate total emissions in 2005 (44.15 $GtCO_2$) we get 29,36 $GtCO_2$. Next, we allocate these energy related emissions to the energy service production sector, fossil fuel extraction and emissions from fertilizer production. From (*45*) we have that 6.4% (2.826 $GtCO_2$ eq) of the total energy related emissions is due to extraction processes. Based on (*46*), fertilizer production is estimated to cause emissions of 0.575 $GtCO_2$ eq. Hence we can split the total energy related emission of 29.36 $GtCO_2$ based on these percentages. This implies that 25,960 $GtCO_2$ will be connected to the energy services

output in our model, 2.826 $GtCO_2$ is attributed to the fossil fuel extraction process and 0.575 $GtCO_2$ is connected to fertilizer production.

The other emission related variables in our model are more straightforward. Emissions from industrial processes in (*45*) are assigned to manufacturing in our model (In total 4.6% = 2.031 $GtCO_2$). Emission from land-use change are assigned to the change in natural land in our model (12.2% = 5.387 $GtCO_2$). Emissions from agriculture are assigned to the total agricultural production variable (13.8% = 6.093 $GtCO_2$). For the fisheries sector, (*47*) estimate carbon dioxide emissions to be approximately 0.14 $GtCO_2$.

Using these assignments as a status quo we can calculate the total policy impact by simply multiplying the percentage change in our model variables resulting from the policy by the status quo emission levels. In total, our model variables cover approximately 97.4% of the emissions outlined in (*45*). The results of this exercise, in terms of percentage changes to each planetary pressure, is outlined in the Table 7 for the carbon tax policy and Table 8 for the combined carbon tax and bio fuel tax policy.

To summarize, we find that a 1% increase in the carbon tax leads to a reduction in carbon dioxide emissions by -0.25% or -0.11 $GtCO_2$ yr^{-1}, which is what we use as an indicator of the change in pressure accrued to the climate change and ocean acidification boundary. For the combination of carbon and biofuel tax, the change is -0.26% or -0.11 $GtCO_2$ yr^{-1}.

Biodiversity loss

Quantitatively assessing loss of biodiversity at a global scale is a notoriously difficult task. Studies that quantify terrestrial biodiversity losses resulting from the environmental pressures of human activities typically focus on land related impacts (*48,49*). There are, however, multiple other environmental pressures causing loss of biodiversity that are not related to land-use (*50*). In (*3*) the global extinction rate is used as one way of quantifying this boundary (defined as

extinctions per million species-years). Here, we will make use of the IUCN Red List of Threatened Species to derive a measure of biodiversity loss. The Red List identifies not only the species that have been confirmed to have gone extinct but also the species that are currently threatened and, if pressures remain, may become extinct in the future. (*51*) identify the drivers behind the prevalent threats to the species on the Red List in a comprehensive assessment of more than 8000 species. These drivers can be directly identified as variables in our model. In (*51*) there is overlap between threats in the sense that multiple activities can pose threats to a given species. We refer to a decrease in an activity posing a threat to a certain number of species as a decrease in threats. Without knowing the overlap between threats, we can not translate this into changes in number of threatened species. Therefore we use the change in threats as our measure. Agricultural activity poses threats to 5295 species, which is the largest number of threats. The second largest threat comes from logging, that threatens 4049 species, and we assign this to timber production in our model. Apart from those, we make the following assignments. Pollution from agriculture threatens 1523 of the species and this is assigned to fertilizer production. Over exploitation (Fishing), threatening 1118 species, is assigned to fisheries production. Energy production (oil and gas) and renewable energy production account for threats to 56 species, which we assign to fossil fuel extraction and renewables. Finally, threats from urban development (industrial), pollution (except agriculture), human disturbance (work), transport, energy production (mining) summed to 3573 which we assign to manufacturing. There are also significant biodiversity effects of climate change, which threatens 1688 of the species. In this analysis we abstract from effects of changes in one ESP on other ESPs (unless the ESP is directly captured by a model variable). We can note, however, that including the effects of climate change would lead to larger decreases of biodiversity loss.

Hence, having connected the categories of threats to species by driver in (*51*) to our model variables, we can measure the biodiversity impact of a policy by assessing whether the number

of threats increases or declines as a result of the policy. For example, if the agricultural production increases by 1% as a result of a policy in our model then this would increase the number of threats from agricultural activity by 52.95 (0.01×5296).[8]

The results, in terms of percentage change to the number of threatened species, are outlined in the column labeled "Biodiv." in Table 7 for the carbon tax policy and Table 8 for the combined carbon tax and biofuel subsidy removal. To summarize, this implies that the total number of threats decline by 0.018% for the carbon tax and by 0.011% for the combined carbon and biofuel tax.

Finally, there are indeed caveats to our approach for assessing biodiversity loss. First, it should be noted that this measure of biodiversity loss is just a proxy for true biodiversity loss. Future work would benefit from assessing the drivers of the actual rate of species loss as defined in e.g. (*1*). Furthermore, we have taken the description of threats in (*51*) and mapped them to our model variables. For instance, all threats assigned to agriculture in (*51*) are assigned to agricultural production in our model. Perhaps some part of these threats come from land use change associated with agriculture rather than agriculture as such. In that case they should be mapped to our land use variables. We do not have a proper basis for such reassignment and therefore stick close to their assignment. Qualitatively, this distinction could matter for the carbon tax in isolation, but will not be important for the carbon tax combined with biofuel policy.

Aerosol loading

Following (*3*) we adopt aerosol optical depth (AOD) as are indicative measure of planetary pressure. To determine how AOD changes as a result of policy we use data from three sources

[8]Clearly this is a crude attempt to capture how biodiversity might respond to changing circumstances, but based on our reading of the literature we have found no better method for assessing this impact on a global scale. It is beyond the scope of this paper to develop such methods which should, however, given the urgency of the problem, be an important area for future research.

(*17, 18, 52*). The impact is calculated as follows. First, we calculate a global average estimate of AOD from the main regional anthropogenic sources (sulfur (0.0392), black carbon (0.0003) and organic carbon (0.0011)) provided by (*18*). Second, we use data from (*17*) to calculate the share of global aerosol contributing emissions for each of these respective sources (sulfur (3.6%), black carbon (32%) and organic carbon (63%)) that stem specifically from biomass burning.[9] Third, using these estimates we can calculate the amount of global AOD which ought to be attributed to emissions from fossil fuel and biofuels (0.038) and biomass burning (0.0022). These estimates are then connected to the model variables fossil fuel consumption (in energy services, fertilizer production and fisheries), biofuel production and change in natural land. In total, a 1% carbon tax leads to a 0.011% decline in AOD and the combined carbon tax and biofuel policy leads to a decline of 0.014% (further details can be found in table 7 and 8) .

Stratospheric ozone depletion and chemical pollution

We do not attempt to quantify the effects on these boundaries. Stratospheric ozone depletion increases with N_2O emissions from agricultural production, fossil fuel use, manufacturing and biofuels. For an increase in the carbon tax all these activities except for biofuel use decreases. The net effect is thus potentially ambiguous. If the carbon tax increase is complemented with a decrease in biofuel subsidies, all relevant variables decrease and we conclude that the net effect is a decrease in the pressure.

Chemical pollution increases in manufacturing, extracted fossil fuels, total agricultural production, agricultural production for food, fossil fuel use in fertilizer production and fossil fuel use in energy services production. All these activities decrease with a carbon tax, with or without a biofuel policy. Hence, we conclude that chemical pollution will decrease in both cases.

[9]These estimates assume that approximately 90% of biomass burning emissions result from land-use change (see (*52*)).

Remaining boundaries

For the remaining boundaries, the impacts are easier to assess since they are directly tied to specific model variables. First, the impact on the biogeochemical flows is assigned to the model variables phosphate and fossil-fuel use in fertilizer production. While the former is self-explanatory, the latter is used as a proxy for nitrogen, which relies almost entirely on fossil fuels in its production. For phosphorus we translate the change into GgPyr^{-1} using the value for current flows (mined and applied to erodible soils) from (*3*): $-0.0001 \times 14000 \approx -1.4$GgPyr^{-1} for the carbon tax and $-0.0005 \times 14000 \approx -7$GgPyr^{-1} for the combined carbon and biofuel policy. For nitrogen we translate the change into TgNyr^{-1} using (*3*): $-0.0014 \times 150 \approx -0.21$TgNyr^{-1} for the carbon tax and $-0.0018 \times 150 \approx -0.27$TgNyr^{-1} for the combined carbon and biofuel policy. Second, for the land-system boundary we rely on the model variable natural land use as an indicator of the direction this boundary is moving in. This is translated into MHa using the average (between high and low value) for "natural forests" in (*31*): $-0.0001 \times 3507 \approx -0.35$MHa for the carbon tax and $0.0004 \times 3507 \approx 1.4$MHa for the combined carbon and biofuel policy. Third, the freshwater boundary is directly tied to the water variable in our model. We translate this into km^{3}yr^{-1} using the value for current use in (*3*): $2600 \times 0.000064 \approx 0.17$km^{3}yr^{-1} for the carbon tax and $-2600 \times 0.00034 \approx -0.88$km^{3}yr^{-1}.

Data and materials availability

All model parameters and data needed to replicate the results in this article are available in the supplementary information. Model code written in Python is available from the authors upon request.

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Competing interests

The authors declare no competing interests.

Table 1: **Earth system processes and their drivers.** The table lists the main pressures on critical ESPs and the economic activities driving them (with percent of total pressure in parenthesis).

ESP	Main pressure	Principal economic drivers
Climate change	CO_2 and other greenhouse gas emissions	Energy prod. (67%), agriculture (14%), land-use change (12%)
Biogeochemical flows	Phosphorus and Nitrogen emissions	Fertilizer use in agriculture (90%)
Ocean acidification	Increased H+ concentration caused by CO_2 emissions	Same as climate change
Freshwater use	Over consumption	Freshwater use in agriculture (92%).
Land-system change	Forest-land loss	Deforestation due to agricultural expansion (96%)
Biodiversity loss	Extinction of species	Agriculture threatens 62%, logging 46%, climate change 19% and fishing 13% of red listed species.†
Stratospheric ozone depletion	Emissions of ozone-depleting substances	Largely resolved, remaining depletion mainly caused by NO_2 emissions from agriculture, fossil fuels, manufacturing and biofuels.
Aerosol loading	Emissions of black and organic carbon, sulfates, nitrates	Fossil and biofuel consumption (95%), biomass burning (5%)†
Chemical pollution	Emissions of non-natural chemicals	Manufacturing, fossil fuels, agriculture and fertilizers.

† Biodiversity loss percentages indicate share of total threatened species by activity. Note that any single species may be subject to more than one threat (hence percentages can sum to more than 100%). Percentage shares of aerosol optical depth contributions for black and organic carbon and sulfur derived from (*17*) and (*18*) (see methods).

Table 2: **Model quantities, prices and uses.**

Variable	*Quantity*	*Price*	*Uses*
A	Agricultural production	p_A	Food $A_{\mathcal{F}}$, biofuels A_B
E	Fossil fuel	p_E	Energy $E_{\mathcal{E}}$, Fisheries E_F and fertilizer production E_P
$\mathcal{E}$	Energy services	$p_{\mathcal{E}}$	Agriculture $\mathcal{E}_A$, manufacturing $\mathcal{E}_Y$
F	Fisheries	p_F	Food
L	Land	p_L	Agriculture L_A, timber production L_T, natural land L_U
M	Other inputs	p_{M_X}	M_X, for $X \in \{A, F, P, T, Y\}$
P	Fertilizers	p_P	Agriculture
$\mathcal{P}$	Phosphate	$p_{\mathcal{P}}$	Fertilizer production
R	Renewables (excluding biofuels)	p_R	Energy services
T	Timber production	p_T	Consumption
W	Fresh water	p_W	Agriculture
Y	Manufacturing	p_Y	Consumption

Table 3: **Parameters - Quantity shares**

Parameter	*source*	*value*
Q_{L,L_A}	source: (*31*)	53%
Q_{L,L_T}	source: (*31*)	2%
$Q_{E,E_{\mathcal{E}}}$	See text	98.2%
Q_{E,E_P}	See text	1.4%
Q_{E,E_F}	See text	0.4%
Q_{A,A_B}	See text	5.15%
$Q_{\mathcal{E},\mathcal{E}_A}$	source: (*32*) page 10	32%

Table 4: **Parameters: expenditure shares**

Expenditure share	*value*	*comment*
$\Gamma^{A}_{L_A}$	19.2%	
$\Gamma^{A}_{\tilde{L}_A}$	80.8%	$1-\Gamma^{A}_{L_A}$
• $\Gamma^{\tilde{L}_A}_{P}$	7.96%	
• $\Gamma^{\tilde{L}_A}_{W}$	2.39%	
• $\Gamma^{\tilde{L}_A}_{\mathcal{E}_A}$	4.12%	
• $\Gamma^{\tilde{L}_A}_{M_A}$	88.0%	$1-\Gamma^{\tilde{L}_A}_{P}-\Gamma^{\tilde{L}_A}_{W}-\Gamma^{\tilde{L}_A}_{\mathcal{E}_A}$
$\Gamma^{\mathcal{E}}_{A_B}$	0.37%	
$\Gamma^{\mathcal{E}}_{E_{\mathcal{E}}}$	94.33%	
$\Gamma^{\mathcal{E}}_{R}$	0.37%	$1-\Gamma^{\mathcal{E}}_{A_B}-\Gamma^{\mathcal{E}}_{E_{\mathcal{E}}}$
$\Gamma^{U}_{\mathcal{F}}$	12.35%	
• $\Gamma^{\mathcal{F}}_{F}$	3.4%	
• $\Gamma^{\mathcal{F}}_{A_{\mathcal{F}}}$	96.6%	$1-\Gamma^{\mathcal{F}}_{F}$
$\Gamma^{U}_{\tilde{\mathcal{F}}}$	87.65%	$1-\Gamma^{U}_{\mathcal{F}}$
• $\Gamma^{\tilde{\mathcal{F}}}_{Y}$	99.10%	
• $\Gamma^{\tilde{\mathcal{F}}}_{L_U}$	0.1711%	
• $\Gamma^{\tilde{\mathcal{F}}}_{T}$	0.7416%	$1-\Gamma^{\tilde{\mathcal{F}}}_{Y}-\Gamma^{\tilde{\mathcal{F}}}_{L_U}$
$\Gamma^{T}_{L_T}$	37.48%	
$\Gamma^{T}_{M_T}$	62.52%	$1-\Gamma^{T}_{L_T}$
$\Gamma^{Y}_{\mathcal{E}_Y}$	6.38%	
$\Gamma^{Y}_{M_Y}$	93.62%	$1-\Gamma^{Y}_{\mathcal{E}_Y}$
$\Gamma^{P}_{E_P}$	10.95%	
$\Gamma^{P}_{\mathcal{P}}$	31.27%	
$\Gamma^{P}_{M_P}$	57.78%	$1-\Gamma^{P}_{E_P}-\Gamma^{P}_{\mathcal{P}}$

Table 5: **Parameters - Elasticities and Quantities.** The values are depicted as [min, max, mean] where the min and max values are used in the sensitivity analysis.

Parameter	*source*	*value*
σ_U	Ref. (*35*)	[0.4 0.6 0.5]
$\sigma_{\mathcal{F}}$	Ref. (*36*).	[1.13, 1.33, 1.23]
σ_F	Standard values for extractive sectors used in GTAP (*37*).	[0.1, 1, 0.2]
σ_T	Standard values for extractive sectors used in GTAP (*37*).	[0.1, 1, 0.2]
$\sigma_{\tilde{\mathcal{F}}}$	Drawn from estimates of substitutability derived from (*38*)	[1.5, 2, 1.8]
σ_A	Ref. (*22*)	[1.10, 1.24, 1.14]
σ_P	Literature suggests high complementarity, see e.g. (*39*).	[0.05 0.3 0.2]
$\sigma_{\tilde{L}_A}$	Assumed based on reading of the literature	[0.25, 0.75, 0.5]
$\sigma_{\mathcal{E}}$	Ref. (*40*)	1.6
σ_Y	Based on Table 2 in (*41*)	[0.1, 1, 0.5]
Λ_R	Ref. (*42*)	1/2.7
Λ_E	Based on estimates in (*43*)	1
Λ_W	Based on (*44*)	1/1.79
$\Lambda_{\mathcal{P}}$	Assumed based on reading of the literature	1/1.5
Λ_{M_X}	Assumed with wide span for robustness	[0, 2, 1]
V_T	Based on (*22*)	0.05
V_A	Based on (*22*)	0.05
τ_E	A global carbon tax currently does not exist.	0

Table 6: **Baseline results**. Percentage change in key model variables from two policy scenarios: (i) a one-percent increase in the carbon tax; (ii) in addition to (i), a one-percent reduction of biofuel subsidies.

	Carbon tax		*Carbon Tax + Biofuel policy*	
Variable	*Quantity*	*Price*	*Quantity*	*Price*
Agricultural Sector: Production				
Total	-0.006	0.032	-0.043	0.001
Biofuels	0.616	0.032	-0.947	0.001
Food	-0.03	0.032	-0.007	0.001
Agricultural Sector: Inputs				
Fertilizer Production	-0.023	0.063	-0.061	0.034
Water Production	0.006	0.004	-0.034	-0.019
Energy For Agriculture	-0.299	0.615	-0.35	0.613
Land-Share Agriculture	0.009	0.018	-0.035	-0.004
Energy-related sectors and Services				
Energy Services	-0.316	0.615	-0.327	0.613
Energy Services in Manufacturing	-0.323	0.615	-0.316	0.613
Renewables Production	0.419	0.155	0.411	0.152
Fossil-Fuel in Energy Services	-0.361	0.643	-0.366	0.638
Fossil-Fuel in Fertilizer Prod.	-0.139	0.643	-0.181	0.638
Fossil-Fuel in Fisheries	-0.214	0.643	-0.221	0.638
Extractive Sectors				
Land-Share Timber	0.002	0.018	0.02	-0.004
Land-Share Natural	-0.011	0.018	0.041	-0.004
Fossil-Fuel Extraction	-0.357	0.643	-0.362	0.638
Phosphate Extraction	-0.009	-0.006	-0.048	-0.032
Fisheries Production	-0.104	0.092	-0.111	0.085
Others				
Timber Production	0.004	0.01	0.017	0.009
Manufacturing	-0.031	0.029	-0.026	0.033

Table 7: **Effects on ESPs – carbon policy.** Change in planetary pressure from a global carbon tax of 1%. Shows the percentage change to our pressure variables for each respective Earth system process resulting from the perturbation of the model variables following the tax. Land-use refers to change in natural land. Ozone depletion and chemical pollution are only assessed qualitatively (-1 denotes a reduction).

	Climate	*Biodiv.*	*Biochem.*	*Freshw.*	*Ocean acid.*	*Land-use*	*Aerosols*	*Ozone*	*Chem.*
Land-Share Agriculture (L_A)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Land-Share Timber (L_T)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Land-Share Other (L_U)	0.0013	0.0	0.0	0.0	0.0013	-0.0109	0.0	0	0
Fossil-Fuel Extracted (E)	-0.0322	-0.0008	0.0	0.0	-0.0322	0.0	0.0	-1	-1
Fossil-Fuel Use Energy Serv. ($E_{\mathcal{E}}$)	-0.2122	0.0	0.0	0.0	-0.2122	0.0	-0.0135	0	0
Fossil-Fuel Use Fertilizer Prod. (E_P)	-0.0018	0.0	-0.1393	0.0	-0.0018	0.0	-0.0001	0	0
Fossil-Fuel Use Fisheries (E_F)	-0.0007	0.0	0.0	0.0	-0.0007	0.0	-0.0	0	0
Agriculture Total Production (A)	-0.0008	-0.0012	0.0	0.0	-0.0008	0.0	0.0	-1	-1
Agriculture Prod. For Biofuels (A_B)	0.0	0.0	0.0	0.0	0.0	0.0	0.0002	1	0
Agriculture Prod. For Food (A_F)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Energy Services ($\mathcal{E}$)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Energy For Agriculture ($\mathcal{E}_A$)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Energy Services For Manufacturing ($\mathcal{E}_Y$)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Fertilizer Production (P)	0.0	-0.0014	0.0	0.0	0.0	0.0	0.0	0	0
Water Production (W)	0.0	0.0	0.0	0.0064	0.0	0.0	0.0	0	0
Phosphate Extraction ($\mathcal{P}$)	0.0	0.0	-0.0095	0.0	0.0	0.0	0.0	0	0
Renewables Production (R)	0.0	0.0009	0.0	0.0	0.0	0.0	0.0	0	0
Fisheries Production (F)	0.0	-0.0045	0.0	0.0	0.0	0.0	0.0	0	0
Timber Production (T)	0.0	0.0006	0.0	0.0	0.0	0.0	0.0	0	0
Manufacturing (Y)	-0.0013	-0.0042	0.0	0.0	-0.0013	0.0	0.0	-1	-1
Other inputs Agriculture. (M_A)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Timber. (M_T)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Fisheries (M_F)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Manufacturing. (M_Y)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Fertilizers (M_P)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Total impact	-0.2477	-0.0105	-0.1488	0.0064	-0.2477	-0.0109	-0.0134	-1	0

Table 8: **Effects on ESPs – carbon and biofuel policy.** Change in planetary pressure from a global carbon and biofuel tax of 1%. Shows the percentage change to our pressure variables for each respective Earth system process resulting from the perturbation of the model variables following the tax. Land-use refers to change in natural land. Ozone depletion and chemical pollution are only assessed qualitatively (-1 denotes a reduction).

	Climate	*Biodiv.*	*Biochem.*	*Freshw.*	*Ocean acid.*	*Land-use*	*Aerosols*	*Ozone*	*Chem.*
Land-Share Agriculture (L_A)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Land-Share Timber (L_T)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Land-Share Other (L_U)	-0.005	0.0	0.0	0.0	-0.005	0.0408	-0.0001	0	0
Fossil-Fuel Extracted (E)	-0.0327	-0.0008	0.0	0.0	-0.0327	0.0	0.0	-1	-1
Fossil-Fuel Use Energy Serv. ($E_{\mathcal{E}}$)	-0.2149	0.0	0.0	0.0	-0.2149	0.0	-0.0137	0	0
Fossil-Fuel Use Fertilizer Prod. (E_P)	-0.0024	0.0	-0.1815	0.0	-0.0024	0.0	-0.0001	0	0
Fossil-Fuel Use Fisheries (E_F)	-0.0007	0.0	0.0	0.0	-0.0007	0.0	-0.0	0	0
Agriculture Total Production (A)	-0.006	-0.0089	0.0	0.0	-0.006	0.0	0.0	-1	-1
Agriculture Prod. For Biofuels (A_B)	0.0	0.0	0.0	0.0	0.0	0.0	-0.0003	-1	0
Agriculture Prod. For Food (A_F)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Energy Services ($\mathcal{E}$)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Energy For Agriculture ($\mathcal{E}_A$)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Energy Services in Manufacturing ($\mathcal{E}_Y$)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Fertilizer Production (P)	0.0	-0.0036	0.0	0.0	0.0	0.0	0.0	0	0
Water Production (W)	0.0	0.0	0.0	-0.0342	0.0	0.0	0.0	0	0
Phosphate Extraction ($\mathcal{P}$)	0.0	0.0	-0.0476	0.0	0.0	0.0	0.0	0	0
Renewables Production (R)	0.0	0.0009	0.0	0.0	0.0	0.0	0.0	0	0
Fisheries Production (F)	0.0	-0.0048	0.0	0.0	0.0	0.0	0.0	0	0
Timber Production (T)	0.0	0.0027	0.0	0.0	0.0	0.0	0.0	0	0
Manufacturing (Y)	-0.0011	-0.0036	0.0	0.0	-0.0011	0.0	0.0	-1	-1
Other inputs Agriculture. (M_A)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Timber. (M_T)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Fisheries (M_F)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Manufacturing. (M_Y)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Other inputs Fertilizers (M_P)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0
Total impact	-0.2627	-0.0181	-0.2291	-0.0342	-0.2627	0.0408	-0.0142	-1	-1